

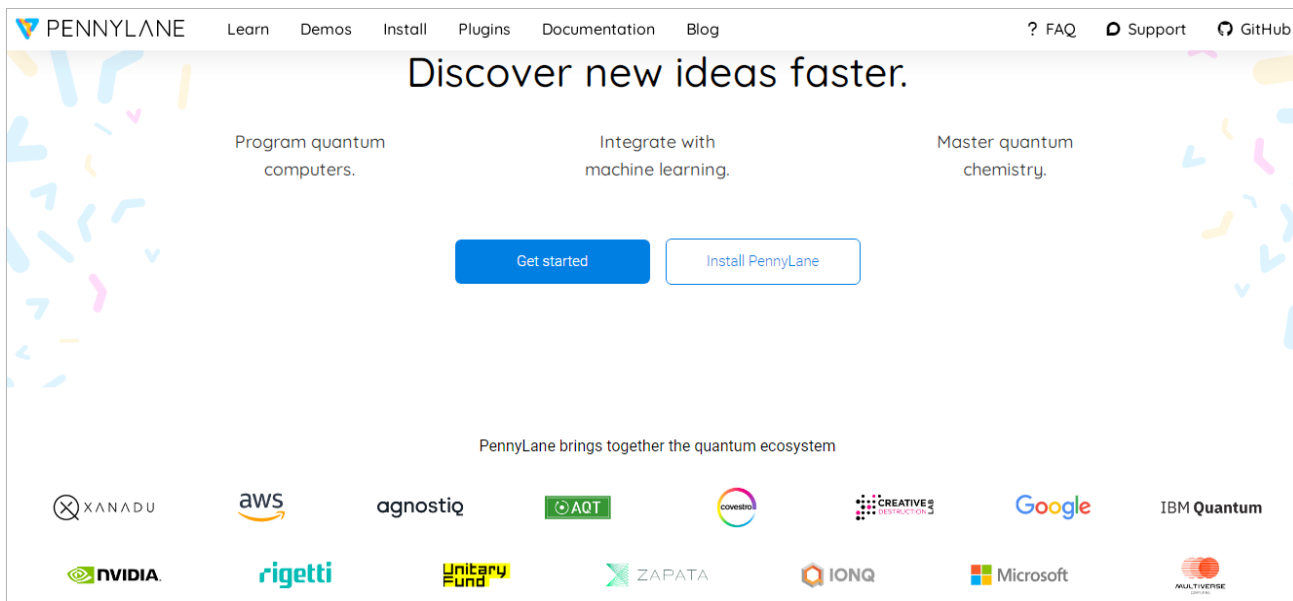


Fig. 2: PennyLane framework for quantum machine learning

calculations, such as in the simulation of chemical reactions and the modelling of financial markets.

**Security.** Quantum computing can potentially offer improved security over classical computing in areas such as cryptography, as certain algorithms that are difficult to break using classical computing can be easily broken by quantum computing.

**Innovative applications.** Quantum computing is a new and rapidly evolving field, with the potential for a wide range of innovative applications in areas such as drug discovery, materials science, and artificial intelligence.

Some of the potential applications of quantum computing are listed below.

- **Cryptography.** Quantum computers have the potential to break many of the cryptographic algorithms that are used to secure sensitive data. However, they can also be used to develop new, more secure encryption methods.

- **Optimisation.** Many real-world optimisation problems, such as supply chain management and logistics, are extremely difficult to solve with classical computers. Quantum computers can provide faster and more efficient solutions to these problems.

- **Machine learning.** Quantum machine learning algorithms can be used to analyse and classify large amounts of data more efficiently than classical algorithms.

- **Chemistry.** Quantum computers can simulate the behaviour of molecules and chemical reactions more accurately than classical computers, which can lead to

the development of new materials and drugs.

- **Finance.** Quantum computing can be used to optimise portfolios, risk assessment, and other financial calculations.

- **Weather forecasting.** Quantum computing can provide more accurate and precise weather forecasts by simulating complex weather patterns and climate models.

- **Particle physics.** Quantum computing can be used to simulate particle interactions, and accelerate the development of new theories and technologies in particle physics.

## Machine learning integration with quantum computing

Quantum machine learning (QML) is a field that explores the intersection between quantum computing and machine learning. It is focused on developing algorithms and techniques that can leverage the unique properties of quantum computing to improve the efficiency and accuracy of machine learning tasks.

Quantum computers use qubits (quantum bits) to perform operations that can solve certain problems exponentially faster than classical computers. This speed can be particularly advantageous for large scale data analysis tasks, such as those encountered in machine learning.

One of the main goals of quantum machine learning is to develop quantum algorithms that outperform classical machine learning algorithms for tasks such as classification, clustering, and regression. Proposed quan-